Atlanta, GA
(703) 953-6963
elliswhunter@gmail.com

Hunter Ellis

Autonomous Systems Engineer

ellishw.tech
github.com/hunterwellis
linkedin.com/in/ellishw

Skills

Software: C++, C, MATLAB, Simulink, Git, ROS2, Gazebo, Make, CMake, Labview, Qt, PyTorch, OpenCV, LaTeX, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino

Hardware: PCB Design and Assembly, Breadboarding, Computer Architecture, Oscilloscope, Multimeter, 3D-Printing

Technical Experience

Autonomous Systems Engineer — **Aug 2025 – Present**
Georgia Tech Research Institute · Applied Systems Lab — *Atlanta, GA*

- Software development for collaborative autonomous UAV swarms.
- Motion planning research for swarm path planning
- Collaborative swarm autonomy
- Software systems integration
- embedded software development
- C++ development
- DDS

Robotics Research / M.S. Computer Engineering — **Aug 2023 – May 2025**
Virginia Tech · Control, Optimization, and Online Learning for Autonomy Lab — *Blacksburg, VA*

- Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

Thrust Vector Control Intern — **May 2024 – Aug 2024**
Jacobs Space Exploration Group · Thrust Vector Control Test Lab — *Huntsville, AL*

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern — **Jun 2023 – Aug 2023**
G2ELab · Power Electronics Lab — *Grenoble, France*

- Researched different inverter control methods for 4-leg (microgrid) topologies in Simulink and HIL simulation.

Naval Research Enterprise Internship Program — **Jun 2022 – Aug 2022**
NSWC Carderock · Center for Innovation in Ship Design — *Bethesda, MD*

- Developed concept hospital sea-train designs and estimated fuel consumption and electrical power loads of the concept sea-trains.

Projects

SLAM FPV Drone Simulator

- Drone simulator for training FPV pilots, written in C++with the Godot game engine.
- Uses SLAM algorithms to create custom environments for pilots to train in.

6-DOF Robot Arm

- 3D printed robot arm, built using stepper motors and pulleys.
- Implemented ROS2 Jazzy control and Gazebo Harmonic simulation.

Education

Master of Science in Computer Engineering — **May 2025**
Virginia Tech – Software and Machine Intelligence — *Blacksburg, VA*

Bachelor of Science in Electrical & Computer Engineering — **May 2024**
Virginia Tech – Control Systems and Machine Learning (double major) — *Blacksburg, VA*